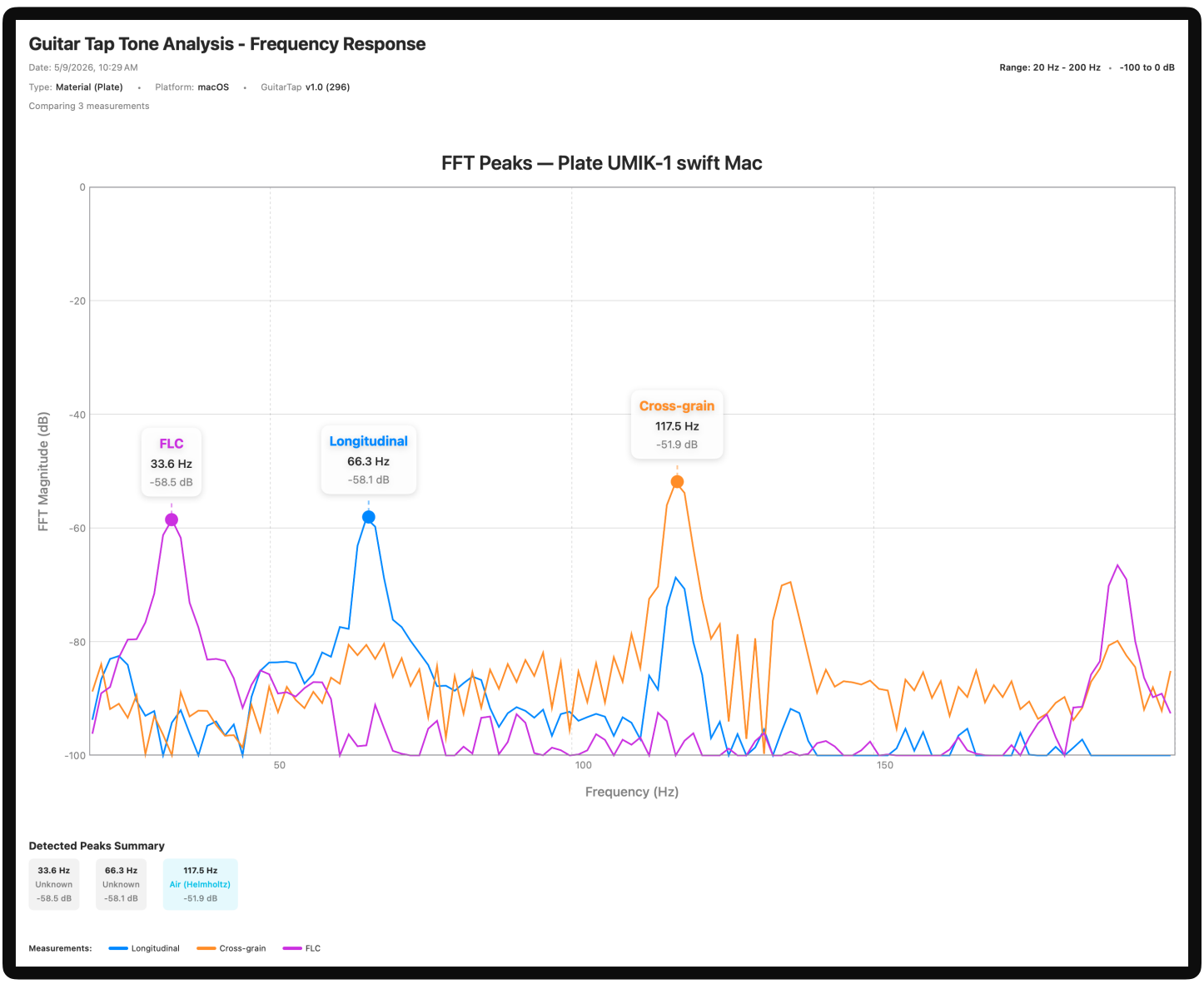
GuitarTap

Tap Tone Analysis Report

May 9, 2026
10:29 AM

Measurement Name:	Plate UMIK-1 swift Mac
Type:	Material (Plate)
Frequency Range:	20.0 Hz – 200.0 Hz
Microphone:	Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
33.6 Hz	-58.5 dB	C1	7.7	FLC (Diagonal)
66.3 Hz	-58.1 dB	C2	15.0	Longitudinal (L)
117.5 Hz	-51.9 dB	A#2	26.7	Cross-grain (C)

Gore Target Thickness

2.91 mm

Body: 490 × 390 mm · f_vs = 60 (Classical Top)
GLC (Shear Modulus): 0.308 GPa

Plate Properties

Sample Dimensions

Length: **557.5 mm** Width: **220.5 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **66.3 Hz** fC: **117.5 Hz** fLC: **33.6 Hz**

Speed of Sound (L): **4134 m/s**
Speed of Sound (C): **1146 m/s**
Young's Modulus (L): **5.96 GPa**
Young's Modulus (C): **0.46 GPa**
GLC (Shear Modulus): **0.308 GPa**

Specific Modulus (L): **17.1** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **11.9**
Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.077**
typical: 0.04 – 0.08

Long/Cross Ratio: **13.0**
typical: 12 – 25

Overall Quality: Good

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.